

Exploiting Cloze Questions for Few Shot Text Classification and Natural Language Inference (Schick and Schütze, 2021)

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Schick and Schütze (2021) introduce Pattern-Exploiting Training (PET), a semi-supervised training procedure that reformulates input examples as cloze-style phrases to help language models understand a given task. Once you have read the paper, consider the following questions.

1. What do you understand by the term few-shot learning? Why is it important /challenging in NLP?
2. What is a Cloze question or Cloze-style phrase?
3. What is pattern-verbalizer pair (PVP)? Give some examples of your own that might be used for different tasks e.g., sentiment classification and paraphrasing
4. How are the PVPs used in training and inference?
5. What is catastrophic forgetting and how is it avoided?
6. How are different PVPs used to create a final model?
7. What is iPET?
8. What experiments did the authors carry out to demonstrate the effectiveness of their approach? What do you think of the results?
9. What does the analysis in section 5 tell us?
10. What are the main conclusions? Are you convinced? Would you use this approach?

References

Timo Schick and Hinrich Schütze. 2021. Exploiting cloze questions for few shot text classification and natural language inference. In *Proceedings of the European Chapter of the Association for Computational Linguistics 2021*.